

PhD Thesis

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Demography and epidemiologi

Incidence rate

Rate

Standardization

age

Lifetable analysis

A lifetable describes how a hypothetical closed birth cohort reduces with rising age alone because of death. The cohort is fictional and consists of ℓ_0 persons, where it is assumed all people are born at the same time. ℓ_0 is called radix and is usually set to 100.000 people. The function ℓ_x (l is used in the examples below) tells us how many people are still alive at the age x , and describes how the table-population declines because of death. As the cohort is closed it must be true that $\ell_{x \geq} \ell_{x+1}$. In the lifetable x is the start of the age interval.

Example of a lifetable

```
# A tibble: 11 × 9
  Alder      l      d      p      q      o      L      T      e
  <fct> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 0      100000 251 0.997 0.00251 1      99774 8334430. 83.3
2 1-9    99749   80 0.999 0.000799 0.997 897385 8234656. 82.6
3 10-19  99670   96 0.999 0.000963 0.997 996216 7337271. 73.6
4 20-29  99574  189 0.998 0.00190 0.996 994789 6341056. 63.7
5 30-39  99384  377 0.996 0.00380 0.994 991955 5346266. 53.8
6 40-49  99007  890 0.991 0.00899 0.990 985620 4354311. 44.0
7 50-59  98117 2821 0.971 0.0288 0.981 967065 3368691. 34.3
8 60-69  95296 7751 0.919 0.0813 0.953 914204 2401626. 25.2
9 70-79  87545 16278 0.814 0.186 0.875 794062 1487422. 17.0
10 80-89 71267 36296 0.491 0.509 0.713 531192 693360. 9.73
11 90+   34971 34971 0      1      0.350 162168 162168. 4.64
```

Table 1: Explanation of the columns in a lifetable

Column	Meaning
Age	Age-interval
l	Decrement function: Number of table persons in the start of the interval
d	Number of deaths in the interval
p	Probability of surviving in the interval given being alive at the start
q	Probability of dying in the interval given being alive at the start
o	Probability of having survived up until the start
L	Total risktime in the interval
T	Total lifetime from the start of the interval
e	Average rest lifetime

From the lifetable example we can see, under the assumption that the mortality rates won't change, that a newborn baby will be expected to live for 83.3 years, and that a 50 year old would be expected to live for another 34.3 years.

$$o_x = \frac{\ell_x}{\ell_0}$$

${}_kq_x$ is the probability for a person at the age x of dying before the age $x + k$. To approximate this probability we use the formular:

$${}_kq_x = \frac{k \cdot {}_kM_x}{1 + (k - {}_ka_x) \cdot {}_kM_x}. \quad (1)$$

This formular depends on the age specific mortality rates ${}_kM_x$, the length of the age interval k and Chiangs a_x . ${}_ka_x$ describes the average lifetime in the age interval for the persons who died in between the ages x and $x + k - 1$. This is usually set to the middle of the interval, i.e. ${}_10a_{50} = 5$. This assumes that the risk of dying in the interval is the same across the ages. This is usually not the case in the first interval and the last.

The central formular can be rewritten to:

$$\frac{k \cdot {}_kM_x}{1 + (k - {}_ka_x) \cdot {}_kM_x} = \frac{k \cdot {}_kD_x}{{}_kR_x + (k - {}_ka_x) \cdot {}_kD_x}$$

$${}_kd_x = {}_kq_x \cdot \ell_x.$$

$$\ell_{x+k} = \ell_x - {}_kd_x.$$

$$\ell_{x+k} = \ell_x(1 - {}_kq_x).$$

$${}_kd_x = \ell_x - \ell_{x+k}.$$

Survival Analysis

Cox model

Breslow Estimator

Parametric